In [7]:
```
# ==========================================================
# 1. EDA NOTEBOOK FOR TEXT AND AUDIO ANALYSIS
# ==========================================================

# ------------------------------------------------
# 1.1 IMPORTS AND SETUP
# ------------------------------------------------
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import os
import librosa
import librosa.display
from collections import Counter
import nltk
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
from textblob import TextBlob
from wordcloud import WordCloud
import IPython.display as ipd
import warnings
warnings.filterwarnings('ignore')

# Download necessary NLTK data
nltk.download('punkt')
nltk.download('stopwords')
nltk.download('wordnet')

# ------------------------------------------------
# 1.2 LOAD AND EXAMINE DATA
# ------------------------------------------------
print("Loading dataset...")
# Use relative path to locate the dataset file
data_path = "data/overview-of-recordings.csv"
data = pd.read_csv(data_path)
print(f"Dataset shape: {data.shape}")
print(f"Dataset columns: {data.columns.tolist()}")

# Display the first few rows
print("\nFirst 5 rows of the dataset:")
print(data.head())

# Basic data information
print("\nBasic information about the dataset:")
print(data.info())
print("\nChecking for missing values:")
print(data.isnull().sum())

# ------------------------------------------------
# 1.3 ANALYZE TARGET VARIABLE (PROMPT DISTRIBUTION)
# ------------------------------------------------
print("\nDistribution of prompt/intent categories:")
prompt_counts = data['prompt'].value_counts()
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print(prompt_counts)

# Plot the distribution of prompts
plt.figure(figsize=(12, 6))
sns.barplot(x=prompt_counts.values, y=prompt_counts.index)
plt.title("Distribution of Medical Prompts")
plt.xlabel("Number of Samples")
plt.tight_layout()
plt.show()

print(f"\nTotal number of samples: {len(data)}")
print(f"Number of unique prompts: {data['prompt'].nunique()}")
print(f"Prompt classes: {', '.join(data['prompt'].unique())}")

# ------------------------------------------------
# 1.4 TEXT DATA EXPLORATION
# ------------------------------------------------
print("\n--- TEXT DATA EXPLORATION ---")

# Basic text statistics
data['text_length'] = data['phrase'].apply(lambda x: len(str(x)) if pd.notna(x) els
data['word_count'] = data['phrase'].apply(lambda x: len(str(x).split()) if pd.notna

print("\nText statistics:")
print(f"Average text length: {data['text_length'].mean():.2f} characters")
print(f"Average word count: {data['word_count'].mean():.2f} words")

# Plot text length distribution
plt.figure(figsize=(16, 6))

plt.subplot(1, 2, 1)
sns.histplot(data['text_length'], kde=True)
plt.title("Distribution of Text Lengths")
plt.xlabel("Number of Characters")

plt.subplot(1, 2, 2)
sns.histplot(data['word_count'], kde=True)
plt.title("Distribution of Word Counts")
plt.xlabel("Number of Words")

plt.tight_layout()
plt.show()

# Text length by prompt category
plt.figure(figsize=(12, 6))
sns.boxplot(x='prompt', y='word_count', data=data)
plt.title("Word Count Distribution by Prompt Category")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

# Text preprocessing function
def preprocess_text(text):
    if pd.isna(text):
        return ""
    # Lowercase
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    text = text.lower()
    # Remove special characters
    text = ''.join([c for c in text if c.isalnum() or c.isspace()])
    # Tokenize
    tokens = word_tokenize(text)
    # Remove stopwords
    stop_words = set(stopwords.words('english'))
    tokens = [token for token in tokens if token not in stop_words]
    # Join back to string
    return ' '.join(tokens)

# Apply preprocessing
data['processed_text'] = data['phrase'].apply(preprocess_text)

# Create word clouds for each prompt class
plt.figure(figsize=(20, 15))
for i, prompt in enumerate(data['prompt'].unique(), 1):
    class_text = ' '.join(data[data['prompt'] == prompt]['processed_text'])
    if class_text.strip():  # Check if text is not empty
        wordcloud = WordCloud(width=400, height=400,
                              background_color='white',
                              max_words=100).generate(class_text)
        plt.subplot(5, 5, i)  # Adjusted for 25 classes
        plt.imshow(wordcloud, interpolation='bilinear')
        plt.title(f"Word Cloud: {prompt}")
        plt.axis('off')
plt.tight_layout()
plt.show()

# Sentiment analysis
data['sentiment'] = data['phrase'].apply(lambda x: TextBlob(str(x)).sentiment.polar
data['subjectivity'] = data['phrase'].apply(lambda x: TextBlob(str(x)).sentiment.su

plt.figure(figsize=(12, 6))
sns.boxplot(x='prompt', y='sentiment', data=data)
plt.title("Sentiment Distribution by Prompt Category")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

plt.figure(figsize=(12, 6))
sns.boxplot(x='prompt', y='subjectivity', data=data)
plt.title("Subjectivity Distribution by Prompt Category")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

# Scatter plot of sentiment vs. subjectivity
plt.figure(figsize=(10, 8))
sns.scatterplot(x='sentiment', y='subjectivity', hue='prompt', data=data)
plt.title("Sentiment vs. Subjectivity by Prompt Category")
plt.xlabel("Sentiment (-1 to 1)")
plt.ylabel("Subjectivity (0 to 1)")
plt.tight_layout()
plt.show()
```

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# ------------------------------------------------
# 1.5 AUDIO DATA EXPLORATION (SAMPLE ANALYSIS)
# ------------------------------------------------
print("\n--- AUDIO DATA EXPLORATION ---")

# Function to extract basic audio features
def extract_audio_features(file_path):
    try:
        # Load audio file
        y, sr = librosa.load(file_path, sr=None)

        # Extract features
        duration = librosa.get_duration(y=y, sr=sr)
        rms = np.mean(librosa.feature.rms(y=y))
        zero_crossing_rate = np.mean(librosa.feature.zero_crossing_rate(y))

        return {
            'duration': duration,
            'rms': rms,
            'zero_crossing_rate': zero_crossing_rate
        }

    except Exception as e:
        print(f"Error extracting features from {file_path}: {e}")
        return None

# Define base audio directory
audio_base_dir = "data/recordings/"

# Function to find audio file in the correct subdirectory
def find_audio_file(file_name):
    # Check each subdirectory
    for subdir in ['train', 'test', 'validate']:
        file_path = os.path.join(audio_base_dir, subdir, file_name)
        if os.path.exists(file_path):
            return file_path, subdir
    return None, None

# Check if audio directory exists
if os.path.exists(audio_base_dir):
    print(f"Audio directory found at {audio_base_dir}")

    # Check subdirectories
    subdirs = [d for d in os.listdir(audio_base_dir)
               if os.path.isdir(os.path.join(audio_base_dir, d))]
    print(f"Found subdirectories: {subdirs}")

    # Sample audio files for analysis
    sample_size = min(50, len(data))
    audio_samples = data.sample(sample_size, random_state=42)

    audio_features = []
    for i, row in audio_samples.iterrows():
        file_name = row['file_name']
        file_path, split = find_audio_file(file_name)
```

```
        if file_path:
            features = extract_audio_features(file_path)
            if features:
                features['file_name'] = file_name
                features['prompt'] = row['prompt']
                audio_features.append(features)

    # Convert to DataFrame
    audio_df = pd.DataFrame(audio_features)

    if not audio_df.empty:
        print(f"Extracted features from {len(audio_df)} audio files")

        # Plot audio feature distributions
        plt.figure(figsize=(16, 12))

        plt.subplot(2, 2, 1)
        sns.histplot(audio_df['duration'], kde=True)
        plt.title("Distribution of Audio Durations")
        plt.xlabel("Duration (seconds)")

        plt.subplot(2, 2, 2)
        sns.boxplot(x='prompt', y='duration', data=audio_df)
        plt.title("Audio Duration by Prompt Category")
        plt.xticks(rotation=45)

        plt.subplot(2, 2, 3)
        sns.scatterplot(x='rms', y='zero_crossing_rate', hue='prompt', data=audio_d
        plt.title("RMS vs. Zero Crossing Rate by Prompt")

        plt.subplot(2, 2, 4)
        sns.boxplot(x='prompt', y='rms', data=audio_df)
        plt.title("RMS Energy by Prompt Category")
        plt.xticks(rotation=45)

        plt.tight_layout()
        plt.show()
    else:
        print("No audio features were successfully extracted")
else:
    print(f"Audio directory not found at {audio_base_dir}")
    print("Audio analysis skipped - check your directory structure")

# ----------------------------------------------
# 1.6 CORRELATION ANALYSIS
# ----------------------------------------------
# Create a correlation matrix of text and audio features (if available)
if 'audio_df' in locals() and not audio_df.empty:
    # Merge audio features with text features
    merged_df = pd.merge(
        data[['file_name', 'text_length', 'word_count', 'sentiment', 'subjectivity'
        audio_df,
        on=['file_name', 'prompt']
    )

    # Select numeric columns for correlation
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```
    numeric_cols = ['text_length', 'word_count', 'sentiment', 'subjectivity',
                    'duration', 'rms', 'zero_crossing_rate']
    corr_matrix = merged_df[numeric_cols].corr()

    # Plot correlation heatmap
    plt.figure(figsize=(10, 8))
    sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', vmin=-1, vmax=1)
    plt.title("Correlation Between Text and Audio Features")
    plt.tight_layout()
    plt.show()

# ------------------------------------------------
# 1.7 ADDITIONAL DATASET FEATURES ANALYSIS
# ------------------------------------------------
# Analyze audio quality features from the dataset
print("\n--- AUDIO QUALITY ANALYSIS ---")

# Look at the distribution of audio quality metrics
quality_features = ['audio_clipping', 'background_noise_audible',
                    'overall_quality_of_the_audio', 'quiet_speaker']

for feature in quality_features:
    if feature in data.columns:
        plt.figure(figsize=(10, 6))
        feature_counts = data[feature].value_counts()
        sns.barplot(x=feature_counts.index, y=feature_counts.values)
        plt.title(f"Distribution of {feature}")
        plt.xticks(rotation=45)
        plt.tight_layout()
        plt.show()

        # Check if quality metrics relate to prompt categories
        if feature != 'overall_quality_of_the_audio':  # Skip numeric features
            plt.figure(figsize=(12, 8))
            cross_tab = pd.crosstab(data['prompt'], data[feature], normalize='index
            cross_tab.plot(kind='bar', stacked=True)
            plt.title(f"Distribution of {feature} by Prompt")
            plt.xticks(rotation=45)
            plt.tight_layout()
            plt.show()

# ------------------------------------------------
# 1.8 SUMMARY STATISTICS AND CONCLUSIONS
# ------------------------------------------------
print("\n--- SUMMARY STATISTICS ---")
print(f"Total samples: {len(data)}")
print(f"Number of unique prompts/intents: {data['prompt'].nunique()}")
print("\nPrompt distribution:")
for prompt, count in prompt_counts.items():
    print(f"  - {prompt}: {count} samples ({count/len(data)*100:.1f}%)")

print("\nText statistics by prompt:")
text_stats = data.groupby('prompt')[['text_length', 'word_count']].agg(['mean', 'st
print(text_stats)

print("\n--- EDA CONCLUSIONS ---")
```

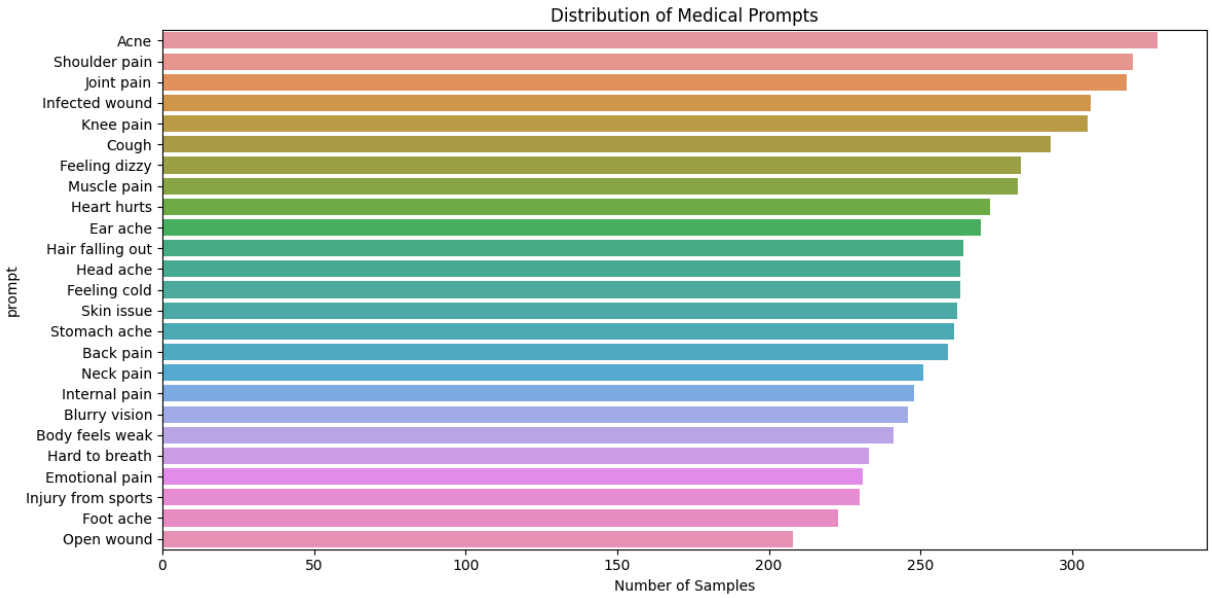
```
print("1. The dataset contains multiple medical prompt categories for multi-class c
print("2. Text length and complexity vary by prompt category")
print("3. Word usage patterns differ significantly between prompt categories")
print("4. Sentiment and subjectivity show variation across different medical condit
if 'audio_df' in locals() and not audio_df.empty:
    print("5. Audio features (duration, energy, etc.) show patterns that may help i
print("6. The dataset is suitable for both text-based and audio-based multi-class c
print("7. Audio quality metrics may provide additional features for classification"
```

```
[nltk_data] Downloading package punkt to C:\Users\S-C-R-E-
[nltk_data]     A-M\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package stopwords to C:\Users\S-C-R-E-
[nltk_data]     A-M\AppData\Roaming\nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
[nltk_data] Downloading package wordnet to C:\Users\S-C-R-E-
[nltk_data]     A-M\AppData\Roaming\nltk_data...
[nltk_data]   Package wordnet is already up-to-date!
```

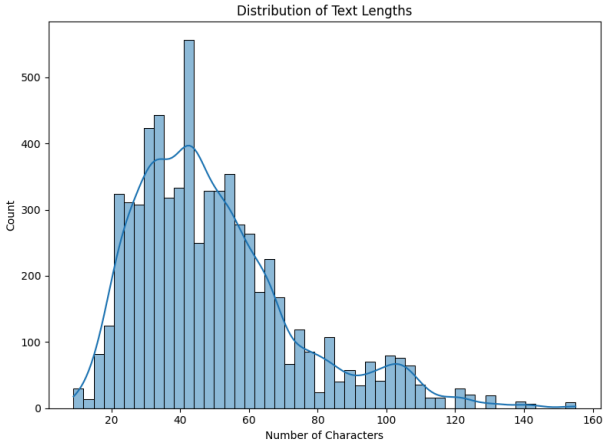
```
Loading dataset...
Dataset shape: (6661, 13)
Dataset columns: ['audio_clipping', 'audio_clipping:confidence', 'background_noise_a
udible', 'background_noise_audible:confidence', 'overall_quality_of_the_audio', 'qui
et_speaker', 'quiet_speaker:confidence', 'speaker_id', 'file_download', 'file_name',
'phrase', 'prompt', 'writer_id']

First 5 rows of the dataset:
   audio_clipping  audio_clipping:confidence background_noise_audible  \
0     no_clipping                     1.0000              light_noise
1  light_clipping                     0.6803                 no_noise
2     no_clipping                     1.0000                 no_noise
3     no_clipping                     1.0000              light_noise
4     no_clipping                     1.0000                 no_noise

   background_noise_audible:confidence  overall_quality_of_the_audio  \
0                               1.0000                          3.33
1                               0.6803                          3.33
2                               0.6655                          3.33
3                               1.0000                          3.33
4                               1.0000                          4.67

     quiet_speaker  quiet_speaker:confidence  speaker_id  \
0  audible_speaker                       1.0    43453425
1  audible_speaker                       1.0    43719934
2  audible_speaker                       1.0    43719934
3  audible_speaker                       1.0    31349958
4  audible_speaker                       1.0    43719934

                                     file_download  \
0  https://ml.sandbox.cf3.us/cgi-bin/index.cgi?do...
1  https://ml.sandbox.cf3.us/cgi-bin/index.cgi?do...
2  https://ml.sandbox.cf3.us/cgi-bin/index.cgi?do...
3  https://ml.sandbox.cf3.us/cgi-bin/index.cgi?do...
4  https://ml.sandbox.cf3.us/cgi-bin/index.cgi?do...

                      file_name  \
0  1249120_43453425_58166571.wav
1  1249120_43719934_43347848.wav
2  1249120_43719934_53187202.wav
3  1249120_31349958_55816195.wav
4  1249120_43719934_82524191.wav

                                           phrase            prompt  \
0                   When I remember her I feel down    Emotional pain
1  When I carry heavy things I feel like breaking...  Hair falling out
2         there is too much pain when i move my arm       Heart hurts
3  My son had his lip pierced and it is swollen a...    Infected wound
4            My muscles in my lower back are aching    Infected wound

   writer_id
0   21665495
1   44088126
2   44292353
3   43755034
4   21665495
```

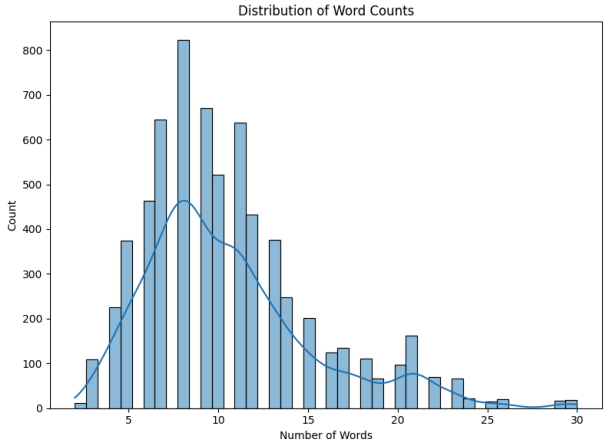
```
Basic information about the dataset:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6661 entries, 0 to 6660
Data columns (total 13 columns):
 #   Column                                Non-Null Count  Dtype
---  ------                                --------------  -----
 0   audio_clipping                        6661 non-null   object
 1   audio_clipping:confidence             6661 non-null   float64
 2   background_noise_audible              6661 non-null   object
 3   background_noise_audible:confidence   6661 non-null   float64
 4   overall_quality_of_the_audio          6661 non-null   float64
 5   quiet_speaker                         6661 non-null   object
 6   quiet_speaker:confidence              6661 non-null   float64
 7   speaker_id                            6661 non-null   int64
 8   file_download                         6661 non-null   object
 9   file_name                             6661 non-null   object
 10  phrase                                6661 non-null   object
 11  prompt                                6661 non-null   object
 12  writer_id                             6661 non-null   int64
dtypes: float64(4), int64(2), object(7)
memory usage: 676.6+ KB
None

Checking for missing values:
audio_clipping                        0
audio_clipping:confidence             0
background_noise_audible              0
background_noise_audible:confidence   0
overall_quality_of_the_audio          0
quiet_speaker                         0
quiet_speaker:confidence              0
speaker_id                            0
file_download                         0
file_name                             0
phrase                                0
prompt                                0
writer_id                             0
dtype: int64

Distribution of prompt/intent categories:
prompt
Acne                  328
Shoulder pain         320
Joint pain            318
Infected wound        306
Knee pain             305
Cough                 293
Feeling dizzy         283
Muscle pain           282
Heart hurts           273
Ear ache              270
Hair falling out      264
Head ache             263
Feeling cold          263
Skin issue            262
```

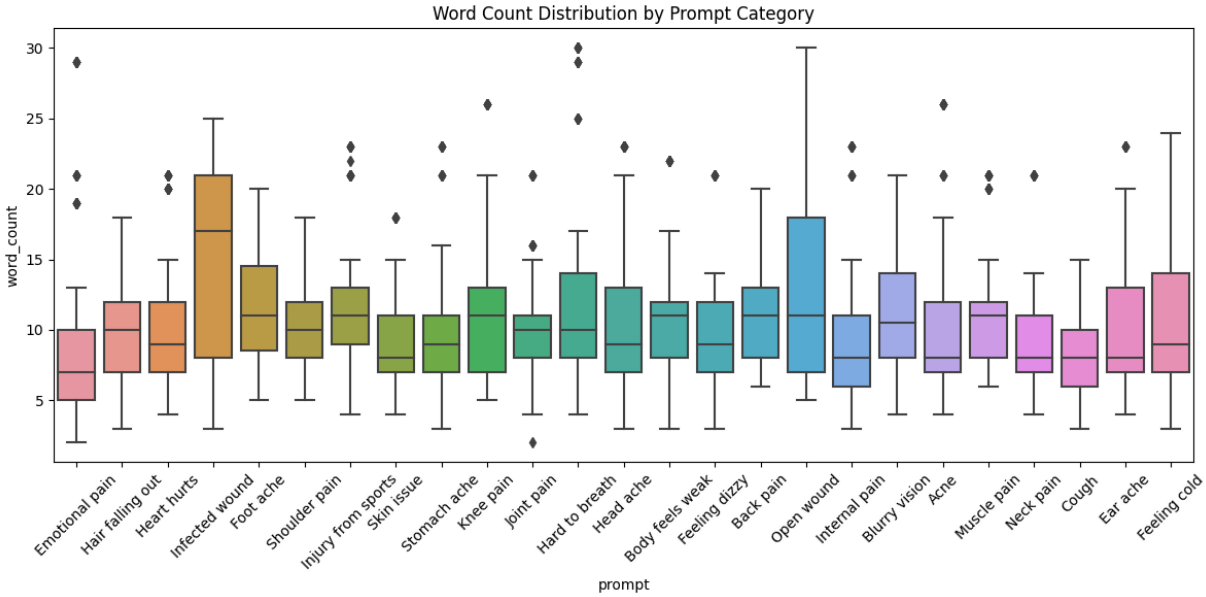
```
Stomach ache          261
Back pain             259
Neck pain             251
Internal pain         248
Blurry vision         246
Body feels weak       241
Hard to breath        233
Emotional pain        231
Injury from sports    230
Foot ache             223
Open wound            208
Name: count, dtype: int64
```



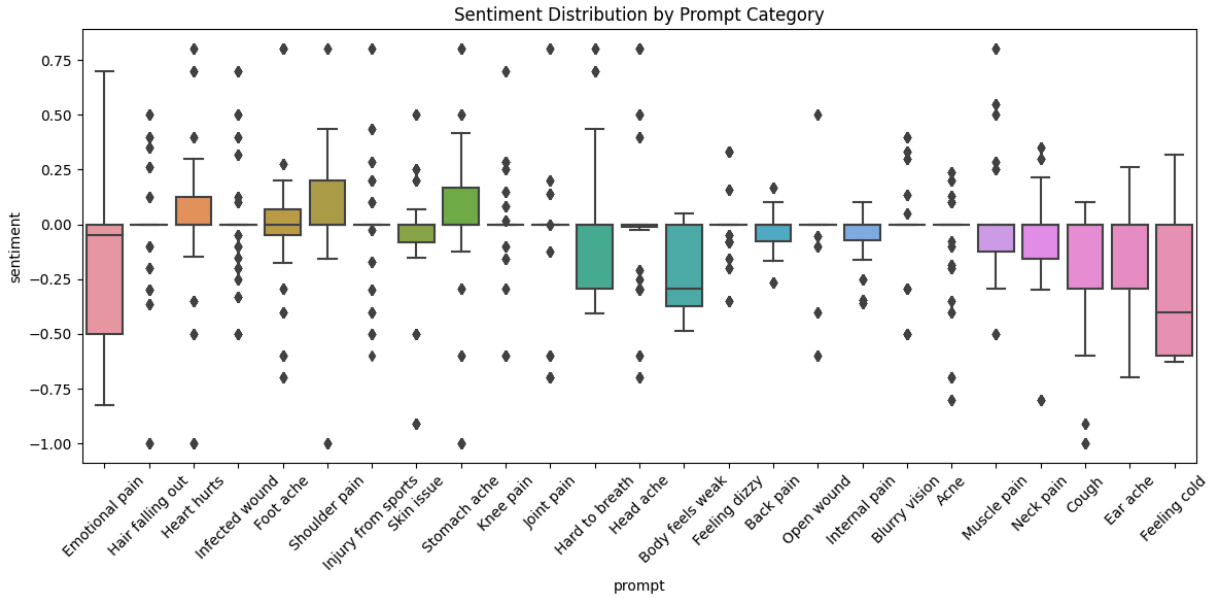

```
Total number of samples: 6661
Number of unique prompts: 25
Prompt classes: Emotional pain, Hair falling out, Heart hurts, Infected wound, Foot
ache, Shoulder pain, Injury from sports, Skin issue, Stomach ache, Knee pain, Joint
pain, Hard to breath, Head ache, Body feels weak, Feeling dizzy, Back pain, Open wou
nd, Internal pain, Blurry vision, Acne, Muscle pain, Neck pain, Cough, Ear ache, Fee
ling cold

--- TEXT DATA EXPLORATION ---

Text statistics:
Average text length: 50.03 characters
Average word count: 10.54 words
```

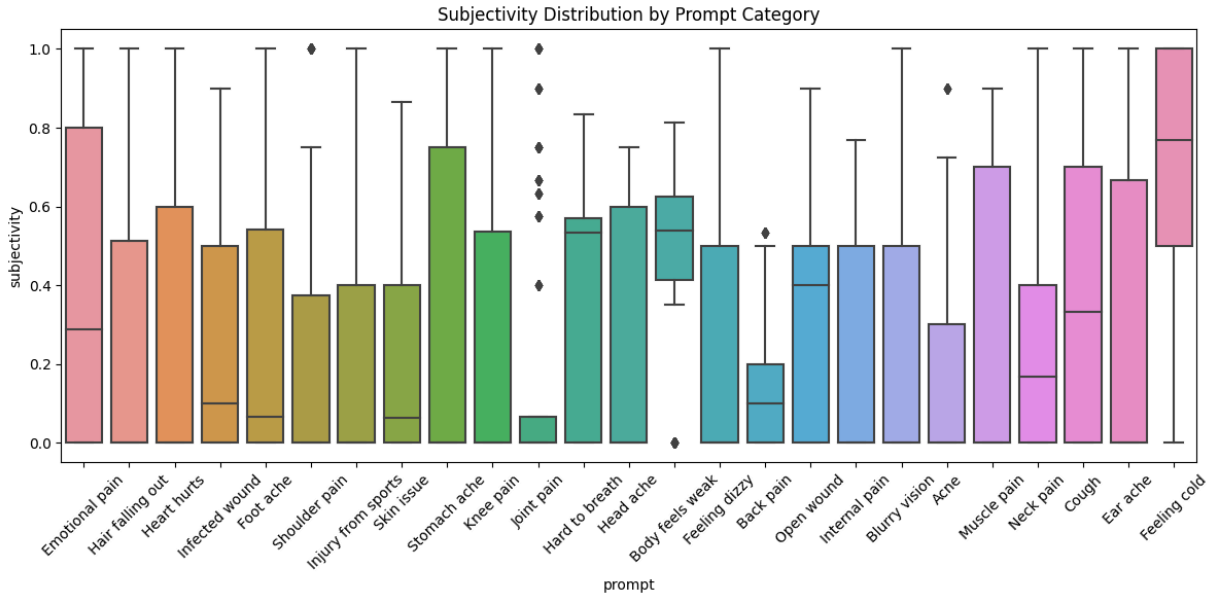

Distribution of Text Lengths
Count
Number of Characters

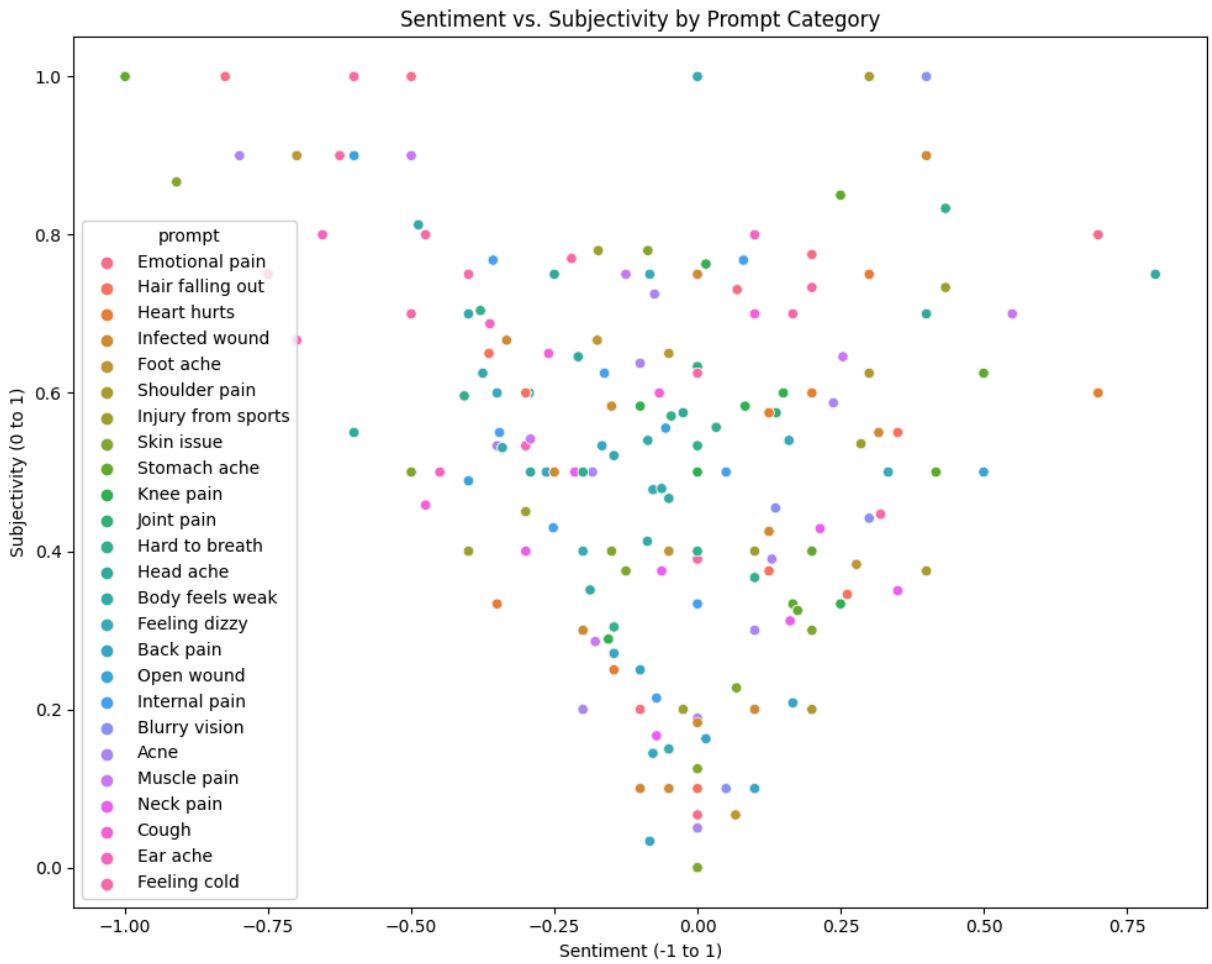

Distribution of Word Counts
Count
Number of Words


Word Count Distribution by Prompt Category
word_count
prompt
Emotional pain
Hair falling out
Heart hurts
Infected wound
Foot ache
Shoulder pain
Injury from sports
Skin issue
Stomach ache
Knee pain
Joint pain
Hard to breath
Head ache
Body feels weak
Feeling dizzy
Back pain
Open wound
Internal pain
Blurry vision
Acne
Muscle pain
Neck pain
Cough
Ear ache
Feeling cold

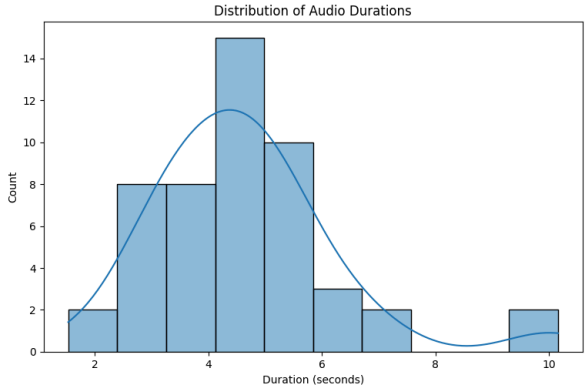
Word Cloud: Emotional pain
Word Cloud: Hair falling out
Word Cloud: Heart hurts
Word Cloud: Infected wound
Word Cloud: Foot ache
Word Cloud: Shoulder pain
Word Cloud: Injury from sports
Word Cloud: Skin issue
Word Cloud: Stomach ache
Word Cloud: Knee pain
Word Cloud: Joint pain
Word Cloud: Hard to breath
Word Cloud: Head ache
Word Cloud: Body feels weak
Word Cloud: Feeling dizzy
Word Cloud: Back pain
Word Cloud: Open wound
Word Cloud: Internal pain
Word Cloud: Blurry vision
Word Cloud: Acne
Word Cloud: Muscle pain
Word Cloud: Neck pain
Word Cloud: Cough
Word Cloud: Ear ache
Word Cloud: Feeling cold

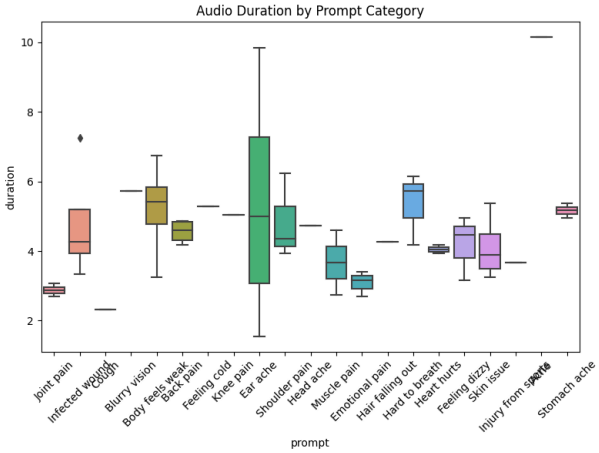

Sentiment Distribution by Prompt Category
sentiment
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
Emotional pain
Hair falling out
Heart hurts
Infected wound
Foot ache
Shoulder pain
Injury from sports
Skin issue
Stomach ache
Knee pain
Joint pain
Hard to breath
Head ache
Body feels weak
Feeling dizzy
Back pain
Open wound
Internal pain
Blurry vision
Acne
Muscle pain
Neck pain
Cough
Ear ache
Feeling cold
prompt

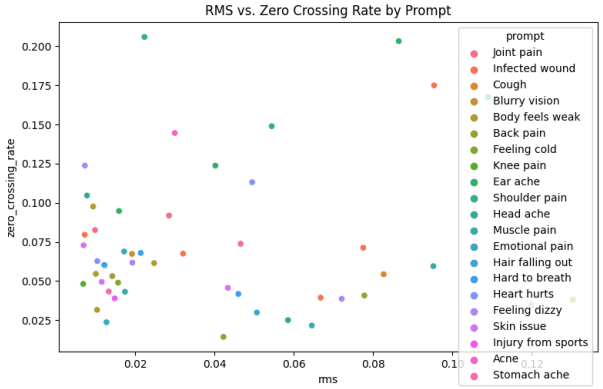


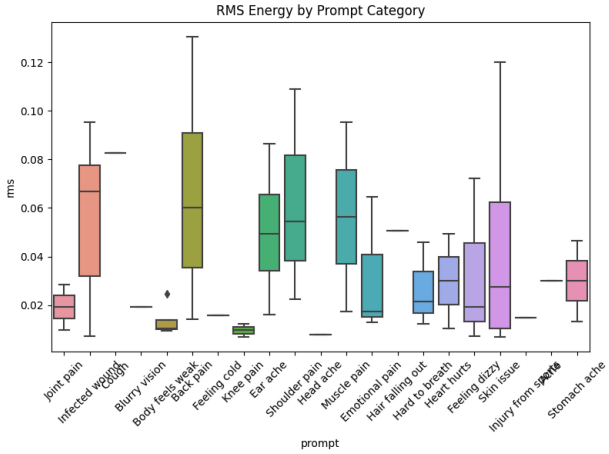


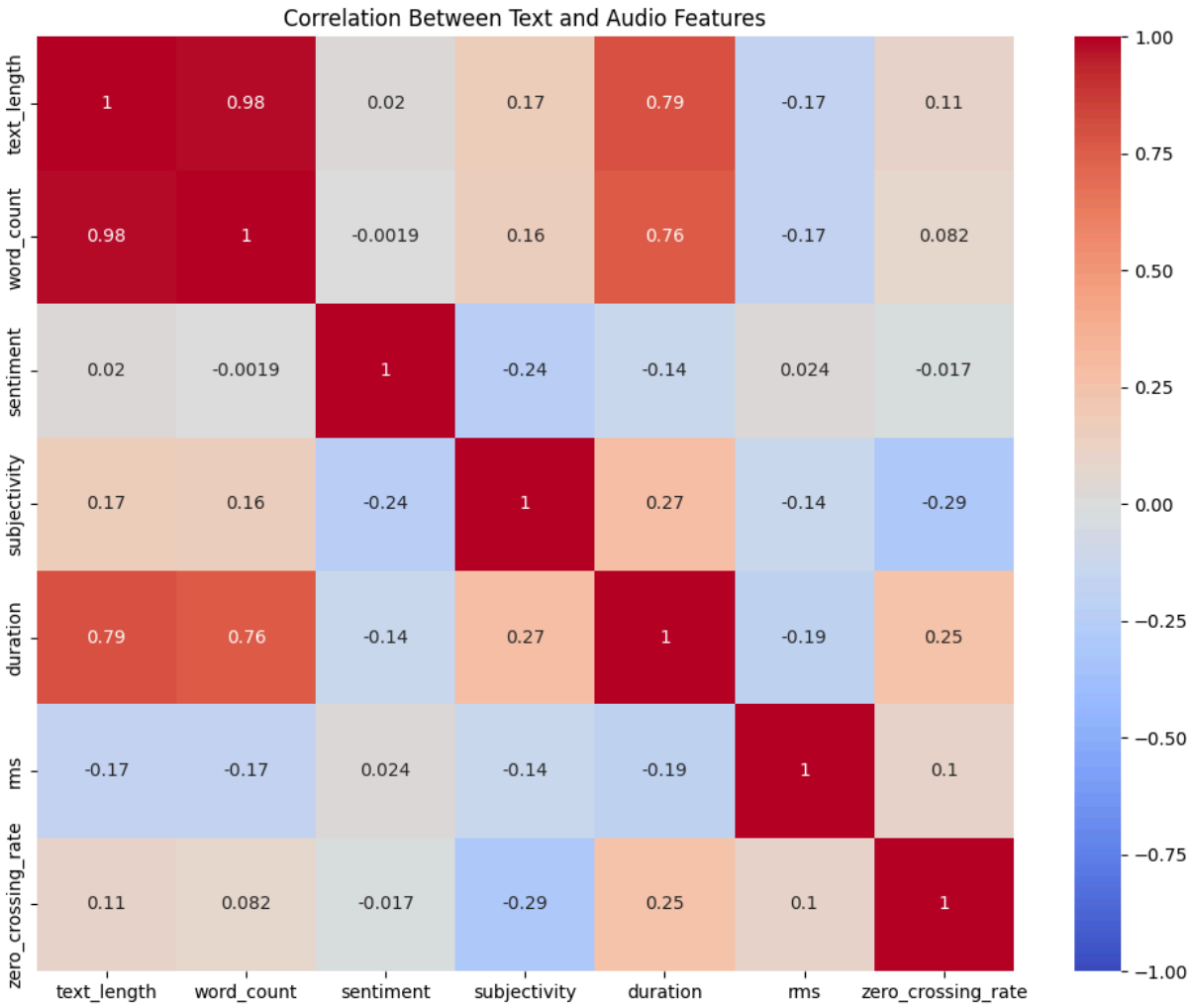

```
--- AUDIO DATA EXPLORATION ---
Audio directory found at data/recordings/
Found subdirectories: ['test', 'train', 'validate']
Extracted features from 50 audio files
```


Distribution of Audio Durations
Count
Duration (seconds)

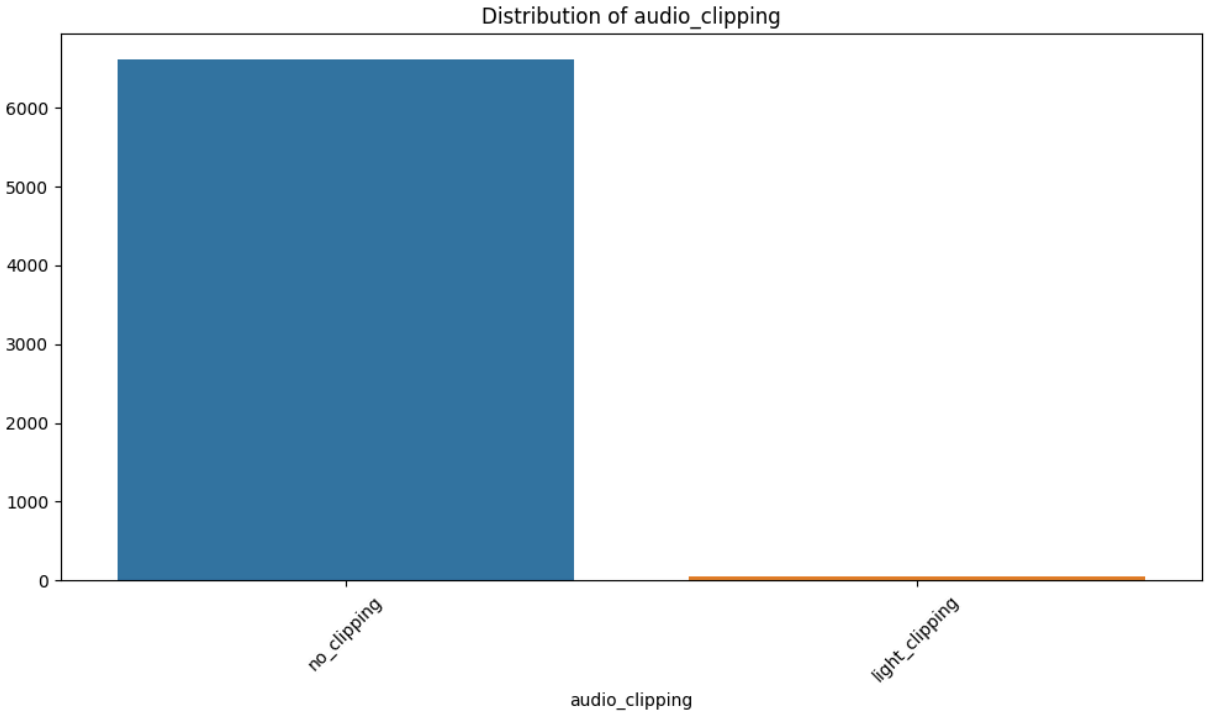

Audio Duration by Prompt Category
duration
prompt
Joint pain
Infected wound
Cough
Blurry vision
Body feels weak
Back pain
Feeling cold
Knee pain
Ear ache
Shoulder pain
Head ache
Muscle pain
Emotional pain
Hair falling out
Hard to breath
Heart hurts
Feeling dizzy
Skin issue
Injury from sports
Acne
Stomach ache


RMS vs. Zero Crossing Rate by Prompt
zero_crossing_rate
rms
prompt
Joint pain
Infected wound
Cough
Blurry vision
Body feels weak
Back pain
Feeling cold
Knee pain
Ear ache
Shoulder pain
Head ache
Muscle pain
Emotional pain
Hair falling out
Hard to breath
Heart hurts
Feeling dizzy
Skin issue
Injury from sports
Acne
Stomach ache

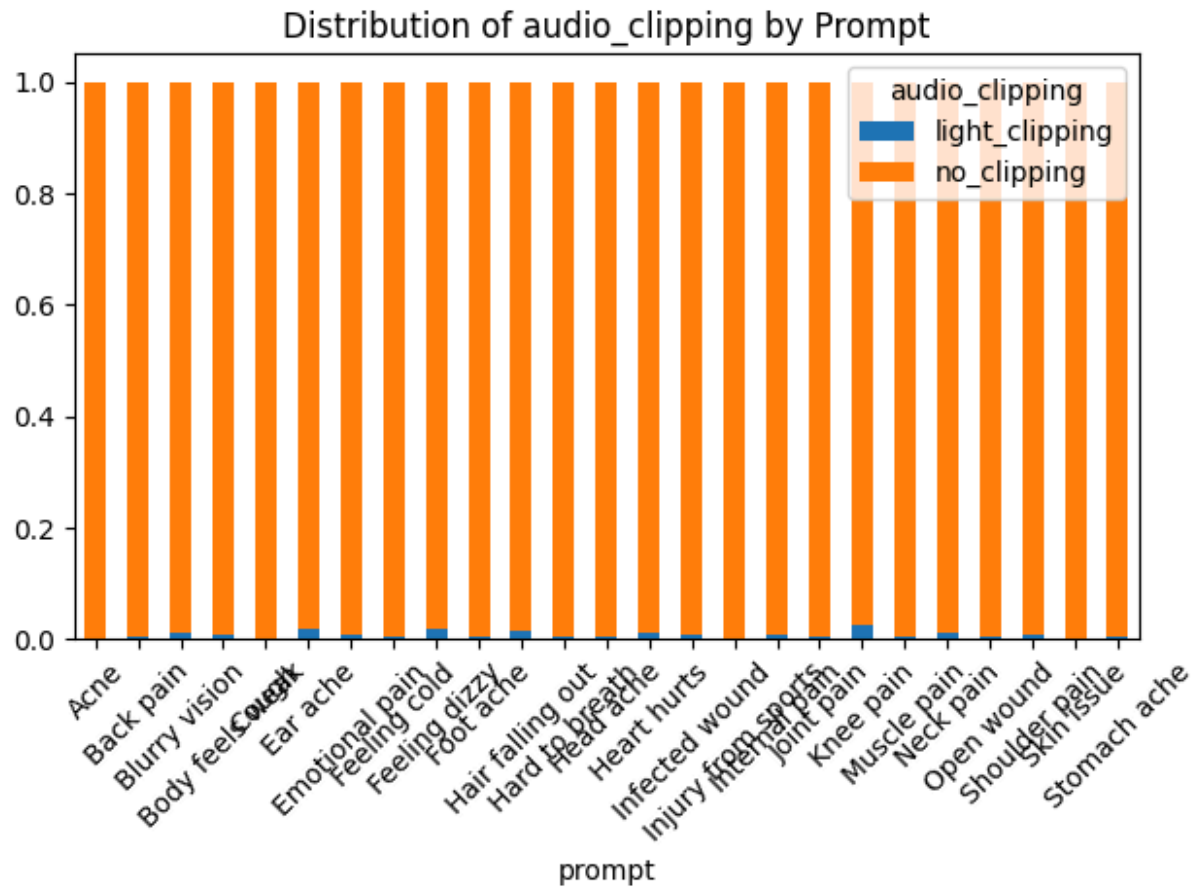

RMS Energy by Prompt Category
rms
prompt
Joint pain
Infected wound
Cough
Blurry vision
Body feels weak
Back pain
Feeling cold
Knee pain
Ear ache
Shoulder pain
Head ache
Muscle pain
Emotional pain
Hair falling out
Hard to breath
Heart hurts
Feeling dizzy
Skin issue
Injury from sports
Acne
Stomach ache

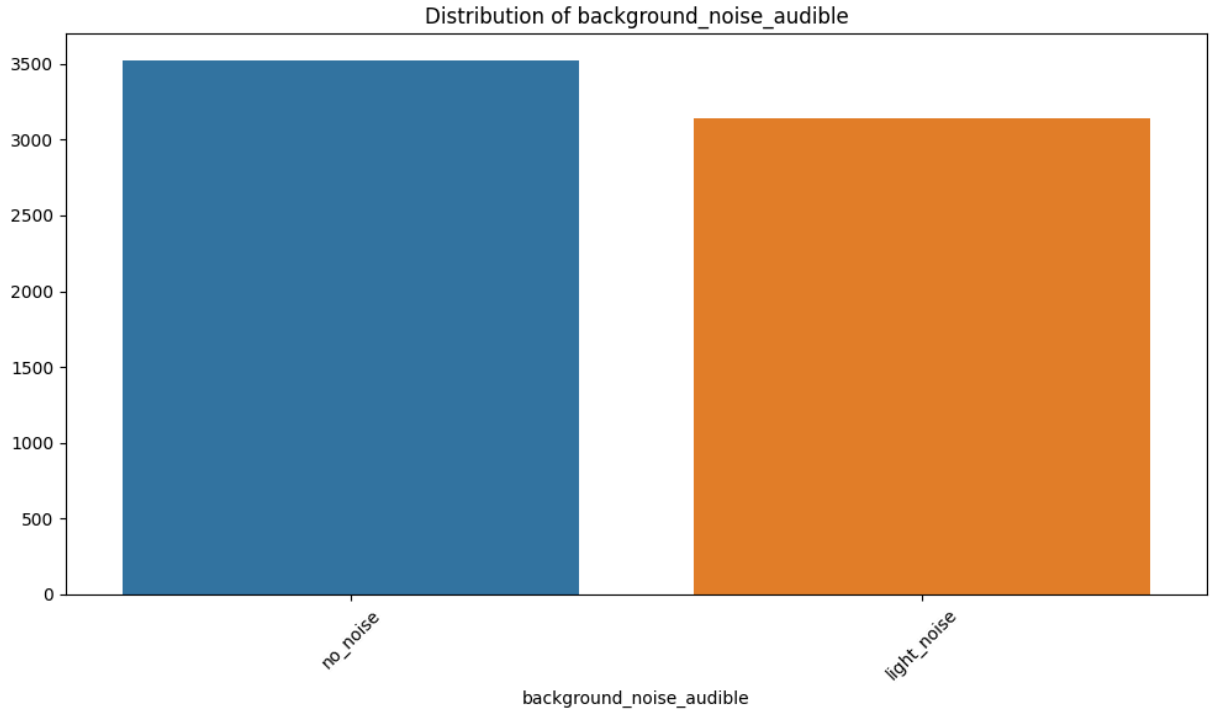



```
--- AUDIO QUALITY ANALYSIS ---
```

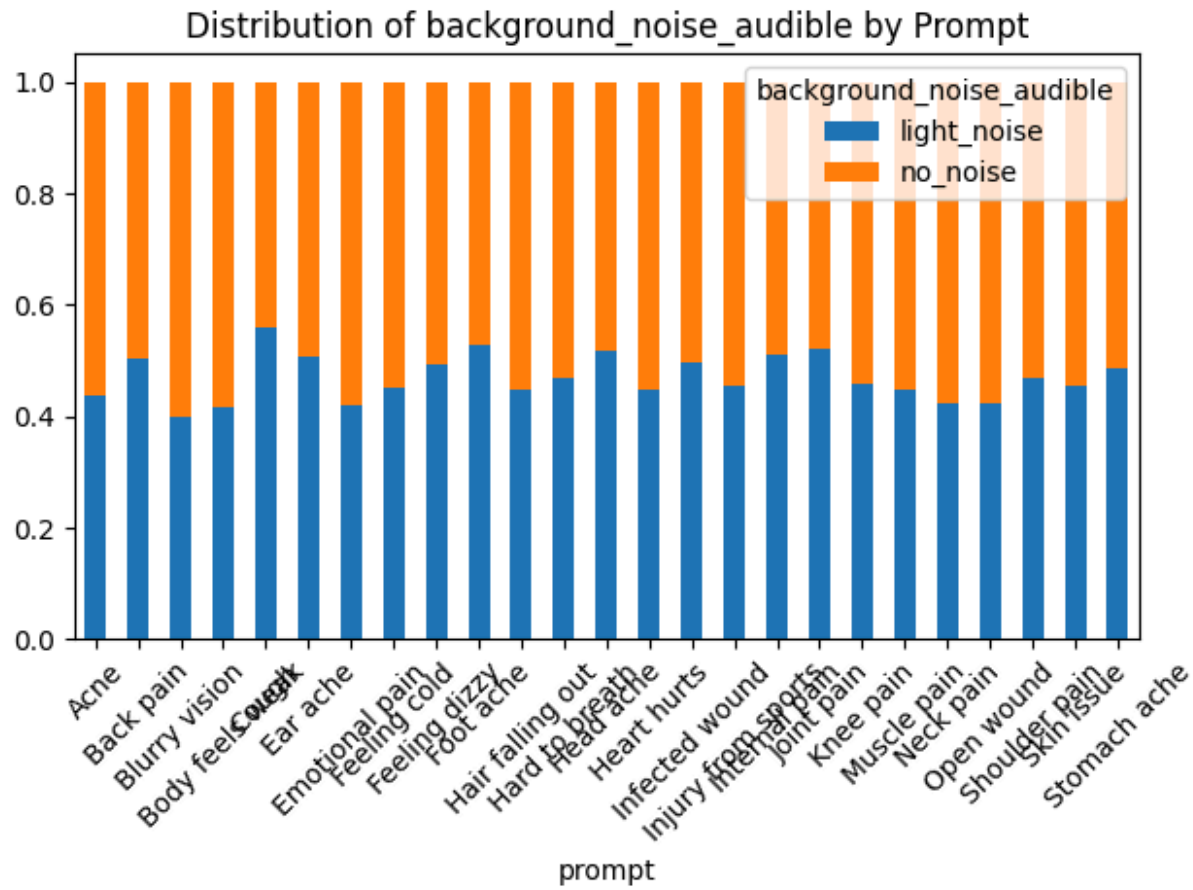


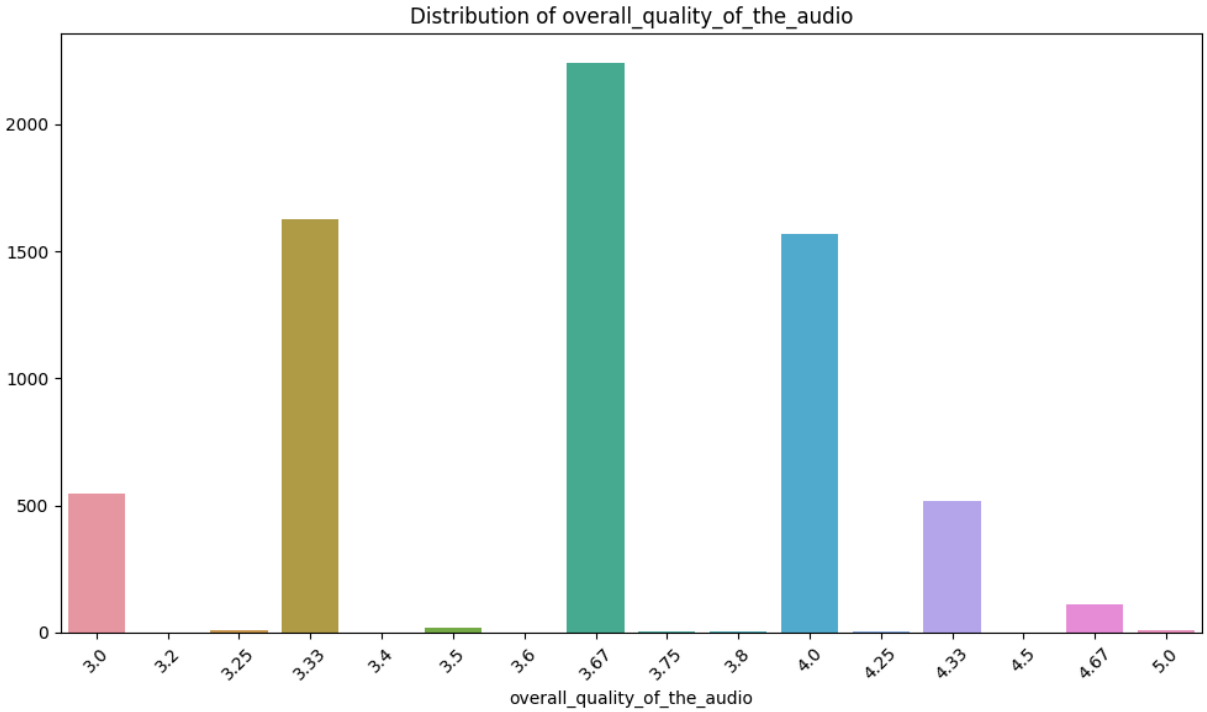

```
<Figure size 1200x800 with 0 Axes>
```

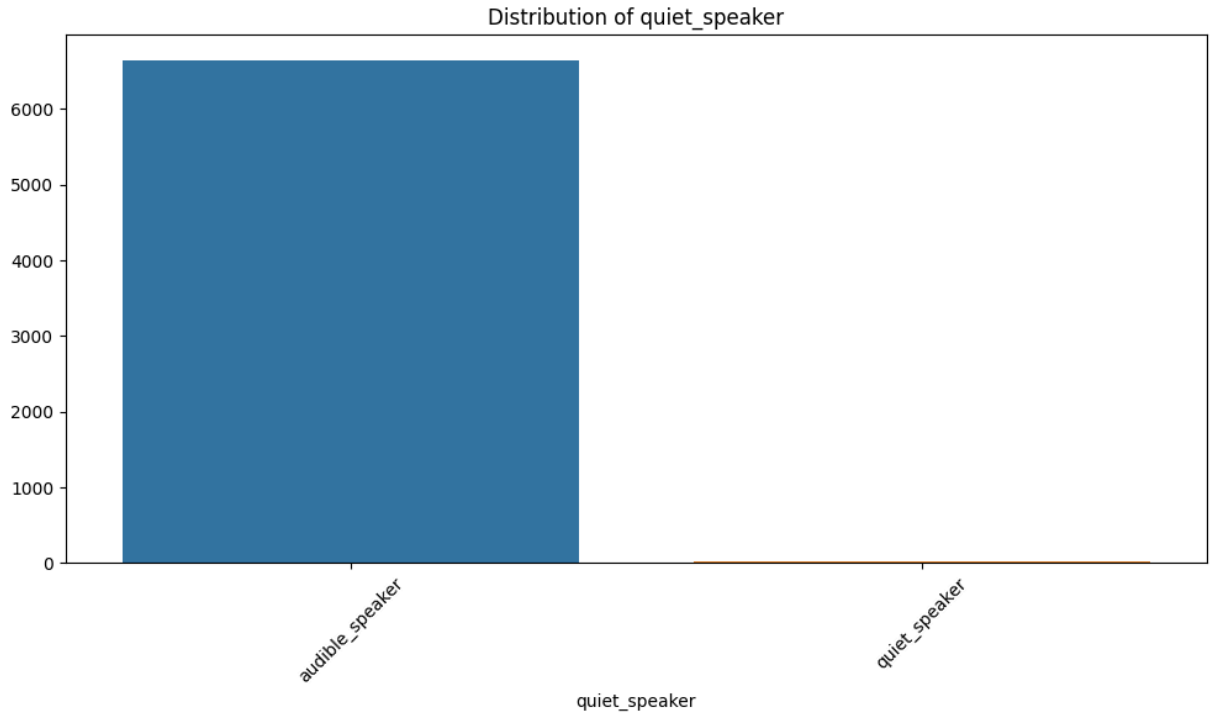




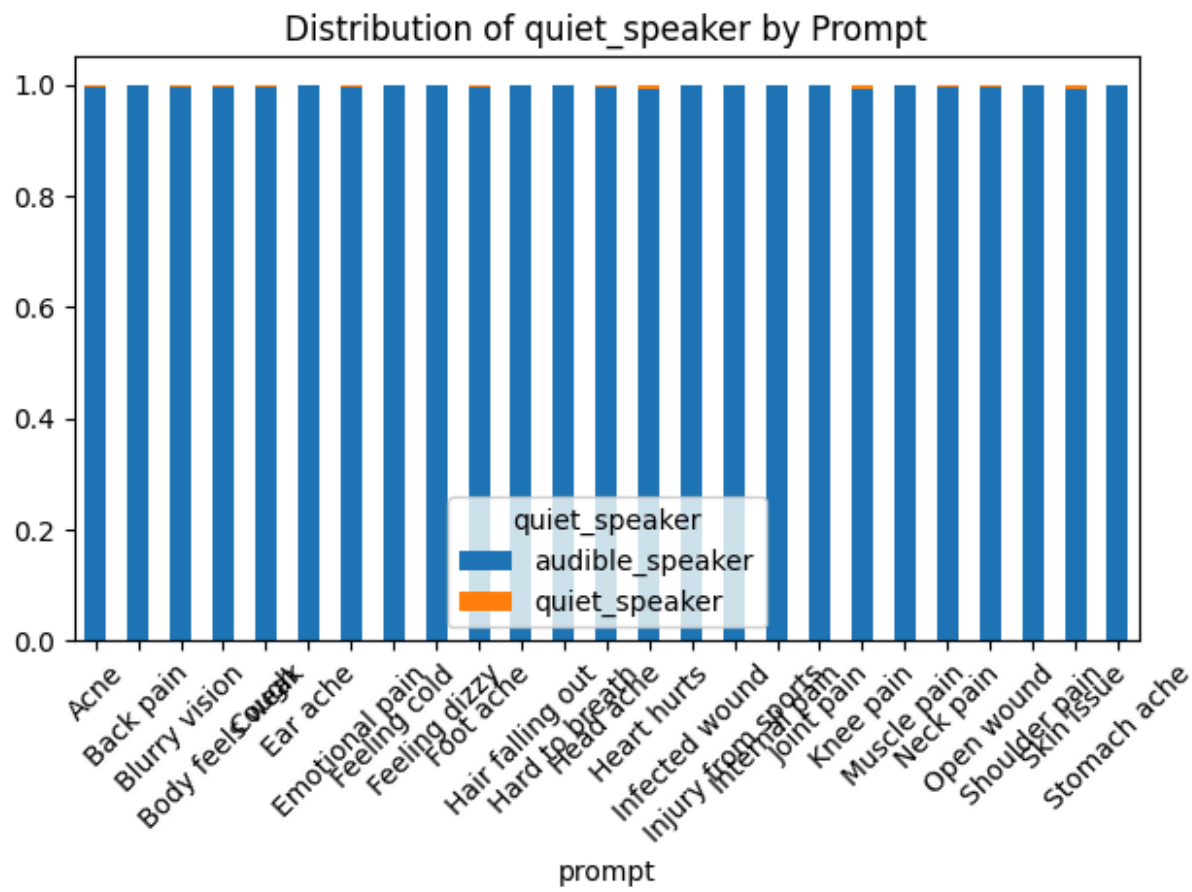

```
<Figure size 1200x800 with 0 Axes>
```


Distribution of background_noise_audible by Prompt
background_noise_audible
light_noise
no_noise
1.0
0.8
0.6
0.4
0.2
0.0
Acne
Back pain
Blurry vision
Body feels weak
Cough
Ear ache
Emotional pain
Feeling cold
Feeling dizzy
Foot ache
Hair falling out
Hard to breath
Head ache
Heart hurts
Infected wound
Injury from sports
Internal pain
Joint pain
Knee pain
Muscle pain
Neck pain
Open wound
Shoulder pain
Skin issue
Stomach ache
prompt


Distribution of overall_quality_of_the_audio
2000
1500
1000
500
0
3.0
3.2
3.25
3.33
3.4
3.5
3.6
3.67
3.75
3.8
4.0
4.25
4.33
4.5
4.67
5.0
overall_quality_of_the_audio




```
<Figure size 1200x800 with 0 Axes>
```



```
--- SUMMARY STATISTICS ---
Total samples: 6661
Number of unique prompts/intents: 25

Prompt distribution:
  - Acne: 328 samples (4.9%)
  - Shoulder pain: 320 samples (4.8%)
  - Joint pain: 318 samples (4.8%)
  - Infected wound: 306 samples (4.6%)
  - Knee pain: 305 samples (4.6%)
  - Cough: 293 samples (4.4%)
  - Feeling dizzy: 283 samples (4.2%)
  - Muscle pain: 282 samples (4.2%)
  - Heart hurts: 273 samples (4.1%)
  - Ear ache: 270 samples (4.1%)
  - Hair falling out: 264 samples (4.0%)
  - Head ache: 263 samples (3.9%)
  - Feeling cold: 263 samples (3.9%)
  - Skin issue: 262 samples (3.9%)
  - Stomach ache: 261 samples (3.9%)
  - Back pain: 259 samples (3.9%)
  - Neck pain: 251 samples (3.8%)
  - Internal pain: 248 samples (3.7%)
  - Blurry vision: 246 samples (3.7%)
  - Body feels weak: 241 samples (3.6%)
  - Hard to breath: 233 samples (3.5%)
  - Emotional pain: 231 samples (3.5%)
  - Injury from sports: 230 samples (3.5%)
  - Foot ache: 223 samples (3.3%)
  - Open wound: 208 samples (3.1%)

Text statistics by prompt:
                   text_length                      word_count                 \
                          mean        std min  max       mean       std min
prompt
Acne                  48.557927  23.393254  15  123   9.939024  4.532850   4
Back pain             49.505792  19.005991  22   95  11.115830  3.853180   6
Blurry vision         52.918699  21.488713  16   99  10.930894  4.381699   4
Body feels weak       49.680498  18.274344  11  106  10.701245  4.011696   3
Cough                 39.194539  14.915046  14   68   7.996587  2.904293   3
Ear ache              49.248148  26.895251  18  112  10.518519  5.191288   4
Emotional pain        44.398268  28.976554  10  131   9.212121  6.375430   2
Feeling cold          54.038023  28.314322  11  123  11.167300  5.478842   3
Feeling dizzy         45.265018  16.829677  18  100   9.632509  3.445514   3
Foot ache             53.668161  22.600869  21  107  11.596413  4.253074   5
Hair falling out      45.549242  17.977104  16   81   9.905303  3.883588   3
Hard to breath        60.248927  31.490390  20  155  12.154506  6.325386   4
Head ache             47.946768  25.119505  16  120  10.463878  4.899527   3
Heart hurts           48.534799  19.959510  14   97  10.413919  4.251396   4
Infected wound        71.738562  31.522414  25  120  14.676471  6.774826   3
Injury from sports    60.460870  23.121773  24  131  11.656522  4.456419   4
Internal pain         44.810484  23.694248  18  109   9.346774  5.018257   3
Joint pain            45.742138  15.427745   9   92  10.069182  3.452461   2
Knee pain             52.737705  23.911915  22  115  11.118033  4.834685   5
Muscle pain           51.418440  18.220342  26   99  11.074468  3.731865   6
Neck pain             42.721116  19.206507  19  106   9.318725  3.741926   4
```

```
Open wound           59.615385  31.406423  23  140  12.596154  6.671765   5
Shoulder pain        49.159375  16.368874  25   85  10.084375  3.222531   5
Skin issue           39.049618  14.931562  17   84   8.835878  3.277663   4
Stomach ache         47.306513  22.745708  20  108   9.632184  4.560992   3


                   max
prompt
Acne                26
Back pain           20
Blurry vision       21
Body feels weak     22
Cough               15
Ear ache            23
Emotional pain      29
Feeling cold        24
Feeling dizzy       21
Foot ache           20
Hair falling out    18
Hard to breath      30
Head ache           23
Heart hurts         21
Infected wound      25
Injury from sports  23
Internal pain       23
Joint pain          21
Knee pain           26
Muscle pain         21
Neck pain           21
Open wound          30
Shoulder pain       18
Skin issue          18
Stomach ache        23

--- EDA CONCLUSIONS ---
1. The dataset contains multiple medical prompt categories for multi-class classific
ation
2. Text length and complexity vary by prompt category
3. Word usage patterns differ significantly between prompt categories
4. Sentiment and subjectivity show variation across different medical conditions
5. Audio features (duration, energy, etc.) show patterns that may help in classifica
tion
6. The dataset is suitable for both text-based and audio-based multi-class classific
ation
7. Audio quality metrics may provide additional features for classification
```